

Q8 Ans: D

Explanation

gravitational force $F = \frac{GMm}{r^2}$

on the ground, $F = W$ and $r = R$: $W = \frac{GMm}{R^2}$ equation 1

at a height $2R$ above the ground, $r = 3R$: $F = \frac{GMm}{(3R)^2}$ equation 2

equation 2 divide by equation 1 gives $F = \frac{W}{9}$

Q8 Ans: B